

```
In [1]:
```

```
import pandas as pd
```

```
In [2]:
```

```
df = pd.read_csv(r'Sample.csv')
```

```
df
```

	Row	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	Customer Name	Segment	Country/Region		City...	Postal Code	Re...
	0	1	US- 2019- 103800	03- 01- 2019	07- 01- 2019	Standard Class	DP-13000	Darren Powers	Consumer	United States	Houston	...77095 Ce	
	1	2	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...60540Ce	
	2	3	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...60540Ce	
	3	4	US- 2019- 112326	04- 01- 2019	08- 01- 2019	Standard Class	PO-19195	Phillina Ober	Home Office	United States	Naperville	...60540Ce	
	4	5	US- 2019- 141817	05- 01- 2019	12- 01- 2019	Standard Class	MB-18085	Mick Brown	Consumer	United States	Philadelphia	...19143 Ea	
	...	...	...	...	...	...	...	...	...	...	...		...
	10189	10190	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell	Consumer	United States	New York City	...10009 Ea	
	10190	10191	US- 2022- 115427	30- 12- 2022	03- 01- 2023	Standard Class	EB-13975	Erica Bern	Corporate	United States	Fairfield	...94533We	
	10191	10192	US- 2022- 156720	30- 12- 2022	03- 01- 2023	Standard Class	JM-15580	Jill Matthias	Consumer	United States	Loveland	...80538We	
	10192	10193	US- 2022- 143259	30- 12- 2022	03- 01- 2023	Standard Class	PO-18865	Patrick O'Donnell	Consumer	United States	New York City	...10009 Ea	
	10193	10194	CA- 2022- 143500	30- 12- 2022	03- 01- 2023	Standard Class	HO-15230	Harry Olson	Consumer	Canada	Charlottetown...	C0A Ea	
10194 rows × 21 columns													

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class
...	...	...	...	...	...
10189	10190	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10190	10191	US-2022-115427	30-12-2022	03-01-2023	Standard Class
10191	10192	US-2022-156720	30-12-2022	03-01-2023	Standard Class
10192	10193	US-2022-143259	30-12-2022	03-01-2023	Standard Class
10193	10194	CA-2022-143500	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
0	DP-13000	Darren Powers	Consumer	United States
1	PO-19195	Phillina Ober	Home Office	United States
2	PO-19195	Phillina Ober	Home Office	United States
3	PO-19195	Phillina Ober	Home Office	United States
4	MB-18085	Mick Brown	Consumer	United States
...	...	...	...	...
10189	PO-18865	Patrick O'Donnell	Consumer	United States
10190	EB-13975	Erica Bern	Corporate	United States
10191	JM-15580	Jill Matthias	Consumer	United States
10192	PO-18865	Patrick O'Donnell	Consumer	United States
10193	HO-15230	Harry Olson	Consumer	Canada

	City	Postal Code	Region	Product ID \
0	Houston	77095	Central	OFF-PA-10000174
1	Naperville	60540	Central	OFF-BI-10004094
2	Naperville	60540	Central	OFF-LA-10003223
3	Naperville	60540	Central	OFF-ST-10002743
4	Philadelphia	19143	East	OFF-AR-10003478
...	...	...	...	...
10189	New York City	10009	East	OFF-BI-10003684
10190	Fairfield	94533	West	OFF-BI-10004632
10191	Loveland	80538	West	OFF-FA-10003472
10192	New York City	10009	East	TEC-PH-10004774
10193	Charlottetown	C0A	East	OFF-BI-10004040

	Category	Sub-Category \
0	Office Supplies	Paper
1	Office Supplies	Binders
2	Office Supplies	Labels
3	Office Supplies	Storage
4	Office Supplies	Art
...	...	...
10189	Office Supplies	Binders
10190	Office Supplies	Binders
10191	Office Supplies	Fasteners
10192	Technology	Phones
10193	Office Supplies	Binders

	Product Name	Sales	Quantity \
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2
1	GBC Standard Plastic Binding Systems Combs	3.540	2
2	Avery 508	11.784	3
3	SAFCO Boltless Steel Shelving	272.736	3
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3
...	...	...	...
10189	Wilson Jones Legal Size Ring Binders	52.776	3
10190	GBC Binding covers	20.720	2
10191	Bagged Rubber Bands	3.024	3
10192	Gear Head AU3700S Headset	90.930	7
10193	Wilson Jones Impact Binders	3.024	3

	Discount	Profit
0	0.2	5.5512
1	0.8	-5.4870
2	0.2	4.2717

```
3         0.2 -64.7748
4         0.2  4.8840
...      ...      ...
10189     0.2 19.7910
10190     0.2  6.4750
10191     0.2 -0.6048
10192     0.0  2.7279
10193     0.2 -0.6048
```

[10194 rows x 21 columns]

In [6]:

```
# 1. Write a function to return total sales only for orders with Profit > 0.
```

```
def total_sales_profit_positive(df):
    return df[df['Profit'] > 0]['Sales'].sum()

print(total_sales_profit_positive(df))
```

stdout:

1826598.7425

In [7]:

```
# 2. Write a function to return the top-selling category (by Sales) in each Region.
```

```
def top_category_region(df):
    sales = df.groupby(['Region', 'Category'])['Sales'].sum().reset_index()
    return sales.loc[sales.groupby('Region')['Sales'].idxmax()]

print(top_category_region(df))
```

stdout:

	Region	Category	Sales
2	Central	Technology	170416.3120
5	East	Technology	267938.0710
8	South	Technology	148771.9080
9	West	Furniture	260679.7295

In [12]:

```
# 3. Create a function to add a column Profit_Ratio = Profit / Sales. Handle division safely.
```

```
def add_profit_ratio(df):
    df['Profit_Ratio'] = df['Profit'] / df['Sales']
    df['Profit_Ratio'] = df['Profit_Ratio'].fillna(0)
    return df

print(add_profit_ratio(df).head())
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	Customer ID	\
0	1	US-2019-103800	03-01-2019	07-01-2019	Standard Class	DP-13000	
1	2	US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
2	3	US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class	PO-19195	
4	5	US-2019-141817	05-01-2019	12-01-2019	Standard Class	MB-18085	

	Customer Name	Segment	Country/Region	City	...	Region	\
0	Darren Powers	Consumer	United States	Houston	...	Central	
1	Phillina Ober	Home Office	United States	Naperville	...	Central	
2	Phillina Ober	Home Office	United States	Naperville	...	Central	
3	Phillina Ober	Home Office	United States	Naperville	...	Central	
4	Mick Brown	Consumer	United States	Philadelphia	...	East	

	Product ID	Category	Sub-Category	\
0	OFF-PA-10000174	Office Supplies	Paper	
1	OFF-BI-10004094	Office Supplies	Binders	
2	OFF-LA-10003223	Office Supplies	Labels	
3	OFF-ST-10002743	Office Supplies	Storage	
4	OFF-AR-10003478	Office Supplies	Art	

	Product Name	Sales	Quantity	\
0	Message Book, Wirebound, Four 5 1/2" X 4" Form...	16.448	2	
1	GBC Standard Plastic Binding Systems Combs	3.540	2	
2	Avery 508	11.784	3	
3	SAFCO Boltless Steel Shelving	272.736	3	
4	Avery Hi-Liter EverBold Pen Style Fluorescent ...	19.536	3	

	Discount	Profit	Profit_Ratio
0	0.2	5.5512	0.3375
1	0.8	-5.4870	-1.5500
2	0.2	4.2717	0.3625
3	0.2	-64.7748	-0.2375
4	0.2	4.8840	0.2500

[5 rows x 22 columns]

In [13]:

```
# 4. Write a function that takes category name and minimum sales as input and returns filtered data.
```

```
def filter_data(df, category_name, min_sales):  
    return df[(df['Category'] == category_name) & (df['Sales'] >= min_sales)]  
  
print(filter_data(df, 'Technology', 100))  
print(filter_data(df, 'Furniture', 100))  
print(filter_data(df, 'Office Supplies', 100))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
12	13	US-2019-167199	06-01-2019	10-01-2019	Standard Class	
13	14	US-2019-167199	06-01-2019	10-01-2019	Standard Class	
31	32	US-2019-162775	13-01-2019	15-01-2019	Second Class	
33	34	US-2019-103366	15-01-2019	17-01-2019	First Class	
37	38	US-2019-115791	16-01-2019	18-01-2019	Second Class	
...	...	...	...	...	...	
10122	10123	US-2022-124296	24-12-2022	28-12-2022	Standard Class	
10142	10143	US-2022-147669	25-12-2022	30-12-2022	Standard Class	
10144	10145	US-2022-105620	25-12-2022	28-12-2022	First Class	
10151	10152	US-2022-134404	27-12-2022	27-12-2022	Same Day	
10182	10183	US-2022-118885	29-12-2022	02-01-2023	Standard Class	
	Customer ID	Customer Name	Segment	Country/Region	\	
12	ME-17320	Maria Etezadi	Home Office	United States		
13	ME-17320	Maria Etezadi	Home Office	United States		
31	CS-12250	Chris Selesnick	Corporate	United States		
33	EH-13990	Erica Hackney	Consumer	United States		
37	DL-13315	Delfina Latchford	Consumer	United States		
...	...	...	...	...		
10122	CS-12355	Christine Sundaresam	Consumer	United States		
10142	SV-20935	Susan Vittorini	Consumer	United States		
10144	JH-15430	Jennifer Halladay	Consumer	United States		
10151	DR-12880	Dan Reichenbach	Corporate	United States		
10182	JG-15160	James Galang	Consumer	United States		
	City	Region	Product ID	Category	Sub-Category	\
12	Henderson	South	TEC-PH-10004539	Technology	Phones	
13	Henderson	South	TEC-PH-10004977	Technology	Phones	
31	Bossier City	South	TEC-AC-10003174	Technology	Accessories	
33	Roswell	South	TEC-AC-10003628	Technology	Accessories	
37	Philadelphia	East	TEC-PH-10004614	Technology	Phones	
...	...	...	...	...	...	
10122	Lafayette	South	TEC-MA-10003183	Technology	Machines	
10142	Fairfield	East	TEC-AC-10002473	Technology	Accessories	
10144	Hialeah	South	TEC-MA-10004626	Technology	Machines	
10151	Newark	East	TEC-PH-10000576	Technology	Phones	
10182	Los Angeles	West	TEC-PH-10002563	Technology	Phones	
	Product Name	Sales Quantity	\			
12	Wireless Extenders zBoost YX545 SOHO Signal Bo...	755.960	4			
13	GE 30524EE4	391.980	2			
31	Plantronics S12 Corded Telephone Headset System	646.740	6			
33	Logitech 910-002974 M325 Wireless Mouse for We...	149.950	5			
37	AT&T 841000 Phone	124.200	3			
...	...	...	...			
10122	DYMO CardScan Personal V9 Business Card Scanner	479.970	3			
10142	Maxell 4.7GB DVD-R	158.928	7			
10144	Lexmark 20R1285 X6650 Wireless All-in-One Printer	120.000	2			
10151	AT&T 1080 Corded phone	164.388	2			
10182	Adtran 1202752G1	302.376	3			
	Discount	Profit	Profit_Ratio			
12	0.0	204.1092	0.270000			
13	0.0	113.6742	0.290000			
31	0.0	258.6960	0.400000			
33	0.0	65.9780	0.440000			
37	0.4	-31.0500	-0.250000			
...	...	...	...			
10122	0.0	239.9850	0.500000			
10142	0.2	41.7186	0.262500			
10144	0.5	-7.2000	-0.060000			
10151	0.4	-35.6174	-0.216667			
10182	0.2	22.6782	0.075000			
[1199 rows x 22 columns]						
	Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
5	6	US-2019-167199	06-01-2019	10-01-2019	Standard Class	
21	22	US-2019-157147	13-01-2019	18-01-2019	Standard Class	
22	23	US-2019-109232	13-01-2019	16-01-2019	Second Class	

34	35	US-2019-115791	16-01-2019	18-01-2019	Second Class
39	40	US-2019-146591	19-01-2019	20-01-2019	First Class
...	...	...	...	...	...
10172	10173	US-2022-158526	29-12-2022	01-01-2023	Second Class
10173	10174	US-2022-118885	29-12-2022	02-01-2023	Standard Class
10174	10175	US-2022-158526	29-12-2022	01-01-2023	Second Class
10175	10176	US-2022-146626	29-12-2022	05-01-2023	Standard Class
10186	10187	US-2022-143259	30-12-2022	03-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
5	ME-17320	Maria Etezadi	Home Office	United States
21	BD-11605	Brian Dahlen	Consumer	United States
22	ND-18370	Natalie DeCherney	Consumer	United States
34	DL-13315	Delfina Latchford	Consumer	United States
39	TS-21340	Toby Swindell	Consumer	United States
...	...	...	...	...
10172	KH-16360	Katherine Hughes	Consumer	United States
10173	JG-15160	James Galang	Consumer	United States
10174	KH-16360	Katherine Hughes	Consumer	United States
10175	BP-11185	Ben Peterman	Corporate	United States
10186	PO-18865	Patrick O'Donnell	Consumer	United States

	City	...	Region	Product ID	Category	Sub-Category \
5	Henderson	...	South	FUR-CH-10004063	Furniture	Chairs
21	San Francisco	...	West	FUR-BO-10003034	Furniture	Bookcases
22	Mount Pleasant	...	South	FUR-CH-10000422	Furniture	Chairs
34	Philadelphia	...	East	FUR-FU-10001095	Furniture	Furnishings
39	Scottsdale	...	West	FUR-BO-10001972	Furniture	Bookcases
...	...	...	...	...	...	...
10172	Louisville	...	South	FUR-CH-10002602	Furniture	Chairs
10173	Los Angeles	...	West	FUR-CH-10002880	Furniture	Chairs
10174	Louisville	...	South	FUR-CH-10004495	Furniture	Chairs
10175	Anaheim	...	West	FUR-FU-10002501	Furniture	Furnishings
10186	New York City	...	East	FUR-BO-10003441	Furniture	Bookcases

	Product Name	Sales	Quantity \
5	Global Deluxe High-Back Manager's Chair	2573.820	9
21	O'Sullivan Elevations Bookcase, Cherry Finish	333.999	3
22	Global Highback Leather Tilter in Burgundy	545.940	6
34	DAX Black Cherry Wood-Tone Poster Frame	127.104	6
39	O'Sullivan 4-Shelf Bookcase in Odessa Pine	181.470	5
...	...	...	...
10172	DMI Arturo Collection Mission-style Design Woo...	1207.840	8
10173	Global High-Back Leather Tilter, Burgundy	393.568	4
10174	Global Leather and Oak Executive Chair, Black	300.980	1
10175	Nu-Dell Executive Frame	101.120	8
10186	Bush Westfield Collection Bookcases, Fully Ass...	323.136	4

	Discount	Profit	Profit_Ratio
5	0.00	746.4078	0.290000
21	0.15	3.9294	0.011765
22	0.00	87.3504	0.160000
34	0.20	28.5984	0.225000
39	0.70	-320.5970	-1.766667
...	...	...	...
10172	0.00	314.0384	0.260000
10173	0.20	-44.2764	-0.112500
10174	0.00	87.2842	0.290000
10175	0.00	37.4144	0.370000
10186	0.20	12.1176	0.037500

[1341 rows x 22 columns]

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
3	4	US-2019-112326	04-01-2019	08-01-2019	Standard Class
8	9	US-2019-167199	06-01-2019	10-01-2019	Standard Class
29	30	US-2019-162775	13-01-2019	15-01-2019	Second Class
30	31	US-2019-157147	13-01-2019	18-01-2019	Standard Class
41	42	US-2019-146591	19-01-2019	20-01-2019	First Class
...	...	...	...	...	...
10149	10150	US-2022-121741	26-12-2022	26-12-2022	Same Day

10168	10169	US-2022-122798	28-12-2022	01-01-2023	Standard Class
10169	10170	US-2022-163979	28-12-2022	02-01-2023	Second Class
10181	10182	US-2022-158673	29-12-2022	04-01-2023	Standard Class
10187	10188	US-2022-126221	30-12-2022	05-01-2023	Standard Class

	Customer ID	Customer Name	Segment	Country/Region	\
3	PO-19195	Phillina Ober	Home Office	United States	
8	ME-17320	Maria Etezadi	Home Office	United States	
29	CS-12250	Chris Selesnick	Corporate	United States	
30	BD-11605	Brian Dahlen	Consumer	United States	
41	TS-21340	Toby Swindell	Consumer	United States	
...	...	...	...	...	...
10149	YC-21895	Yoseph Carroll	Corporate	United States	
10168	SV-20935	Susan Vittorini	Consumer	United States	
10169	KH-16690	Kristen Hastings	Corporate	United States	
10181	KB-16600	Ken Brennan	Corporate	United States	
10187	CC-12430	Chuck Clark	Home Office	United States	

	City	...	Region	Product ID	Category	\
3	Naperville	...	Central	OFF-ST-10002743	Office Supplies	
8	Henderson	...	South	OFF-BI-10004632	Office Supplies	
29	Bossier City	...	South	OFF-ST-10000025	Office Supplies	
30	San Francisco	...	West	OFF-ST-10000078	Office Supplies	
41	Scottsdale	...	West	OFF-EN-10002504	Office Supplies	
...	...	...	...	...	...	...
10149	Fremont	...	Central	OFF-ST-10004459	Office Supplies	
10168	Albuquerque	...	West	OFF-ST-10003058	Office Supplies	
10169	San Francisco	...	West	OFF-ST-10003208	Office Supplies	
10181	Grand Rapids	...	Central	OFF-PA-10000994	Office Supplies	
10187	Columbus	...	Central	OFF-AP-10002457	Office Supplies	

	Sub-Category	Product Name	\
3	Storage	SAFCO Boltless Steel Shelving	
8	Binders	Ibico Hi-Tech Manual Binding System	
29	Storage	Fellowes Stor/Drawer Steel Plus Storage Drawers	
30	Storage	Tennsco 6- and 18-Compartment Lockers	
41	Envelopes	Tyvek Top-Opening Peel & Seel Envelopes, Plai...	
...	...	...	...
10149	Storage	Tennsco Single-Tier Lockers	
10168	Storage	Eldon Mobile Mega Data Cart Mega Stackable A...	
10169	Storage	Adjustable Depth Letter/Legal Cart	
10181	Paper	Xerox 1915	
10187	Appliances	Eureka The Boss Plus 12-Amp Hard Box Upright V...	

	Sales	Quantity	Discount	Profit	Profit_Ratio
3	272.736	3	0.2	-64.7748	-0.2375
8	609.980	2	0.0	274.4910	0.4500
29	572.580	6	0.0	34.3548	0.0600
30	1325.850	5	0.0	238.6530	0.1800
41	108.720	5	0.2	36.6930	0.3375
...	...	...	...	...	...
10149	750.680	2	0.0	37.5340	0.0500
10168	118.250	5	0.0	34.2925	0.2900
10169	725.840	4	0.0	210.4936	0.2900
10181	209.700	2	0.0	100.6560	0.4800
10187	209.300	2	0.0	56.5110	0.2700

[1279 rows x 22 columns]

In [14]:

5. Write a function to find duplicate rows based on Order ID and Product Name.

```
def find_duplicates(df):
    return df[df.duplicated(subset=['Order ID', 'Product Name'])]

print(find_duplicates(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
391	392	US-2019-150119	23-04-2019	27-04-2019	Standard Class
1699	1700	CA-2019-153623	24-11-2019	05-12-2019	Standard Class
2019	2020	CA-2019-123625	29-12-2019	02-01-2020	Standard Class
2022	2023	CA-2019-123625	29-12-2019	02-01-2020	Standard Class
2880	2881	US-2020-103135	24-07-2020	28-07-2020	Standard Class
4465	4466	US-2021-140571	15-03-2021	19-03-2021	Standard Class
4617	4618	US-2021-123750	15-04-2021	21-04-2021	Standard Class
5526	5527	US-2021-129714	01-09-2021	03-09-2021	First Class
6723	6724	US-2021-137043	23-12-2021	25-12-2021	Second Class
9385	9386	US-2022-152912	09-11-2022	12-11-2022	Second Class
9801	9802	US-2022-118017	03-12-2022	06-12-2022	Second Class

	Customer ID	Customer Name	Segment	Country/Region	City \
391	LB-16795	Laurel Beltran	Home Office	United States	Columbus
1699	JP-11135	James Peterman	Corporate	Canada	St. John's
2019	BG-18435	Bruce Galang	Consumer	Canada	Quebec City
2022	BG-18435	Bruce Galang	Consumer	Canada	Quebec City
2880	SS-20515	Shirley Schmidt	Home Office	United States	Louisville
4465	SJ-20125	Sanjit Jacobs	Home Office	United States	Jackson
4617	RB-19795	Ross Baird	Home Office	United States	Gastonia
5526	AB-10060	Adam Bellavance	Home Office	United States	New York City
6723	LC-17140	Logan Currie	Consumer	United States	Springfield
9385	BM-11650	Brian Moss	Corporate	United States	Columbia
9801	LC-16870	Lena Cacioppo	Consumer	United States	Thornton

	...	Region	Product ID	Category	Sub-Category \
391	...	East	FUR-CH-10002965	Furniture	Chairs
1699	...	East	FUR-FU-10002501	Furniture	Furnishings
2019	...	East	FUR-FU-10004093	Furniture	Furnishings
2022	...	East	OFF-FA-10000089	Office Supplies	Fasteners
2880	...	South	OFF-BI-10000069	Office Supplies	Binders
4465	...	South	OFF-PA-10001954	Office Supplies	Paper
4617	...	South	TEC-AC-10004659	Technology	Accessories
5526	...	East	OFF-PA-10001970	Office Supplies	Paper
6723	...	South	FUR-FU-10003664	Furniture	Furnishings
9385	...	East	OFF-ST-10003208	Office Supplies	Storage
9801	...	West	TEC-AC-10002006	Technology	Accessories

	Product Name	Sales	Quantity \
391	Global Leather Highback Executive Chair with P...	281.372	2
1699	Nu-Dell Executive Frame	99.120	8
2019	Hand-Finished Solid Wood Document Frame	68.460	2
2022	Acco Glide Clips	19.600	5
2880	GBC Prepunched Paper, 19-Hole, for Binding Sys...	90.060	6
4465	Xerox 1964	45.680	2
4617	Imation Secure+ Hardware Encrypted USB 2.0 Fla...	291.960	5
5526	Xerox 1881	49.120	4
6723	Electrix Architect's Clamp-On Swing Arm Lamp, ...	286.380	3
9385	Adjustable Depth Letter/Legal Cart	544.380	3
9801	Memorex Micro Travel Drive 16 GB	102.336	8

	Discount	Profit	Profit_Ratio
391	0.3	-12.0588	-0.042857
1699	0.0	35.4144	0.357288
2019	0.0	20.5380	0.300000
2022	0.0	9.6040	0.490000
2880	0.0	41.4276	0.460000
4465	0.0	21.0128	0.460000
4617	0.2	54.7425	0.187500
5526	0.0	23.0864	0.470000
6723	0.0	83.0502	0.290000
9385	0.0	157.8702	0.290000
9801	0.2	14.0712	0.137500

[11 rows x 22 columns]

```

In [16]:

# 6. Write a function to return monthly total sales using Order Date.

def monthly_sales(df):
    df['Order Date'] = pd.to_datetime(df['Order Date'], dayfirst=True)
    return df.groupby(df['Order Date'].dt.to_period('M'))['Sales'].sum()

print(monthly_sales(df))

```

```

stdout:
Order Date
2019-01    14518.0550
2019-02     4519.8920
2019-03    56933.9090
2019-04    28295.3450
2019-05    26319.7670
2019-06    34669.4796
2019-07    33946.3930
2019-08    28918.3385
2019-09    82670.4288
2019-10    32413.3390
2019-11    78826.9567
2019-12    72008.3085
2020-01    18461.9156
2020-02    11951.4110
2020-03    39978.6280
2020-04    34195.2085
2020-05    30246.9465
2020-06    24797.2920
2020-07    28765.3250
2020-08    37057.0042
2020-09    64627.0460
2020-10    31407.1335
2020-11    75972.5635
2020-12    75532.5572
2021-01    18830.3310
2021-02    22978.8150
2021-03    53031.0690
2021-04    38829.1590
2021-05    57042.8380
2021-06    40937.1480
2021-07    40300.4990
2021-08    31716.8143
2021-09    73521.6629
2021-10    59831.0010
2021-11    79411.9658
2021-12    97502.2770
2022-01    44259.2140
2022-02    20301.1334
2022-03    60728.4808
2022-04    36779.0361
2022-05    45155.4822
2022-06    53056.0777
2022-07    45989.4960
2022-08    64129.7580
2022-09    88064.5320
2022-10    83474.7832
2022-11    118454.5050
2022-12     85175.0328
Freq: M, Name: Sales, dtype: float64

```

In [17]:

7. Write a function to sort data first by Region and then by Profit descending.

```
def sort_region_profit(df):  
    return df.sort_values(by=['Region', 'Profit'], ascending=[True, False])  
  
print(sort_region_profit(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode \
5894	5895	US-2021-118689	2021-10-02	09-10-2021	Standard Class
6674	6675	US-2021-117121	2021-12-17	21-12-2021	Standard Class
1203	1204	US-2019-116904	2019-09-23	28-09-2019	Standard Class
6882	6883	US-2022-138289	2022-01-16	18-01-2022	Second Class
4116	4117	US-2020-128587	2020-12-24	30-12-2020	Standard Class
...	...	...	...	...	...
1985	1986	US-2019-148383	2019-12-27	31-12-2019	Standard Class
4100	4101	US-2020-103471	2020-12-24	28-12-2020	Standard Class
4664	4665	US-2021-109869	2021-04-22	29-04-2021	Standard Class
5913	5914	US-2021-157490	2021-10-06	07-10-2021	First Class
7441	7442	US-2022-134845	2022-04-17	23-04-2022	Standard Class

	Customer ID	Customer Name	Segment	Country/Region \
5894	TC-20980	Tamara Chand	Corporate	United States
6674	AB-10105	Adrian Barton	Consumer	United States
1203	SC-20095	Sanjit Chand	Consumer	United States
6882	AR-10540	Andy Reiter	Consumer	United States
4116	HM-14860	Harry Marie	Corporate	United States
...	...	...	...	...
1985	RP-19390	Resi Polking	Consumer	United States
4100	JR-15670	Jim Radford	Consumer	United States
4664	TN-21040	Tanja Norvell	Home Office	United States
5913	LB-16795	Laurel Beltran	Home Office	United States
7441	SR-20425	Sharelle Roach	Home Office	United States

	City	...	Region	Product ID	Category \
5894	Lafayette	...	Central	TEC-CO-10004722	Technology
6674	Detroit	...	Central	OFF-BI-10000545	Office Supplies
1203	Minneapolis	...	Central	OFF-BI-10001120	Office Supplies
6882	Jackson	...	Central	OFF-BI-10004995	Office Supplies
4116	Springfield	...	Central	TEC-CO-10003763	Technology
...	...	...	...	...	...
1985	Phoenix	...	West	OFF-BI-10003650	Office Supplies
4100	Colorado Springs	...	West	FUR-BO-10002613	Furniture
4664	Phoenix	...	West	FUR-TA-10001889	Furniture
5913	Pueblo	...	West	TEC-MA-10001695	Technology
7441	Louisville	...	West	TEC-MA-10000822	Technology

	Sub-Category	Product Name \
5894	Copiers	Canon imageCLASS 2200 Advanced Copier
6674	Binders	GBC Ibimaster 500 Manual ProClick Binding System
1203	Binders	Ibico EPK-21 Electric Binding System
6882	Binders	GBC DocuBind P400 Electric Binding System
4116	Copiers	Canon PC1060 Personal Laser Copier
...	...	...
1985	Binders	GBC DocuBind 300 Electric Binding Machine
4100	Bookcases	Atlantic Metals Mobile 4-Shelf Bookcases, Cust...
4664	Tables	Bush Advantage Collection Racetrack Conference...
5913	Machines	Zebra GK420t Direct Thermal/Thermal Transfer P...
7441	Machines	Lexmark MX611dhe Monochrome Laser Printer

	Sales	Quantity	Discount	Profit	Profit_Ratio
5894	17499.950	5	0.0	8399.9760	0.480000
6674	9892.740	13	0.0	4946.3700	0.500000
1203	9449.950	5	0.0	4630.4755	0.490000
6882	5443.960	4	0.0	2504.2216	0.460000
4116	4899.930	7	0.0	2302.9671	0.470000
...	...	...	...	...	...
1985	946.764	6	0.7	-694.2936	-0.733333
4100	590.058	7	0.7	-786.7440	-1.333333
4664	1272.630	6	0.5	-814.4832	-0.640000
5913	703.710	6	0.7	-938.2800	-1.333333
7441	2549.985	5	0.7	-3399.9800	-1.333333

[10194 rows x 22 columns]

```
In [18]:  
  
# 8. Write a function to return rows where Sales > (mean + 2 * std).  
  
def high_sales(df):  
    threshold = df['Sales'].mean() + 2 * df['Sales'].std()  
    return df[df['Sales'] > threshold]  
  
print(high_sales(df))
```

stdout:

	Row ID	Order ID	Order Date	Ship Date	Ship Mode	\
5	6	US-2019-167199	2019-01-06	10-01-2019	Standard Class	
199	200	US-2019-148838	2019-03-17	21-03-2019	Standard Class	
224	225	US-2019-131275	2019-03-18	24-03-2019	Standard Class	
226	227	US-2019-145317	2019-03-18	23-03-2019	Standard Class	
245	246	US-2019-163223	2019-03-21	25-03-2019	Standard Class	
...	...	...	...	...	...	
9921	9922	US-2022-148446	2022-12-10	14-12-2022	Second Class	
10027	10028	US-2022-115602	2022-12-18	24-12-2022	Standard Class	
10047	10048	US-2022-158729	2022-12-19	21-12-2022	First Class	
10066	10067	US-2022-120376	2022-12-22	25-12-2022	First Class	
10121	10122	US-2022-145219	2022-12-24	25-12-2022	First Class	

	Customer ID	Customer Name	Segment	Country/Region	\
5	ME-17320	Maria Etezadi	Home Office	United States	
199	CP-12340	Christine Phan	Corporate	United States	
224	SC-20050	Sample Company A	Home Office	United States	
226	SM-20320	Sean Miller	Home Office	United States	
245	KH-16690	Kristen Hastings	Corporate	United States	
...	...	...	...	...	
9921	MC-17845	Michael Chen	Consumer	United States	
10027	DJ-13630	Doug Jacobs	Consumer	United States	
10047	AC-10450	Amy Cox	Consumer	United States	
10066	TP-21130	Theone Pippenger	Consumer	United States	
10121	RM-19675	Robert Marley	Home Office	United States	

	City	...	Region	Product ID	Category	\
5	Henderson	...	South	FUR-CH-10004063	Furniture	
199	New York City	...	East	FUR-TA-10003473	Furniture	
224	Burbank	...	West	OFF-ST-10000078	Office Supplies	
226	Jacksonville	...	South	TEC-MA-10002412	Technology	
245	Springfield	...	South	TEC-PH-10000730	Technology	
...	...	...	...	...	...	
9921	North Las Vegas	...	West	FUR-TA-10004256	Furniture	
10027	New York City	...	East	OFF-AP-10000891	Office Supplies	
10047	Lafayette	...	South	OFF-SU-10002881	Office Supplies	
10066	Detroit	...	Central	FUR-CH-10002335	Furniture	
10121	Los Angeles	...	West	TEC-CO-10001449	Technology	

	Sub-Category	Product Name	\
5	Chairs	Global Deluxe High-Back Manager's Chair	
199	Tables	Bretford Rectangular Conference Table Tops	
224	Storage	Tennsco 6- and 18-Compartment Lockers	
226	Machines	Cisco TelePresence System EX90 Videoconferenci...	
245	Phones	Samsung Galaxy S4 Active	
...	...	...	
9921	Tables	Bretford ♦Just In Time♦ Height-Adjustable Mult...	
10027	Appliances	Kensington 7 Outlet MasterPiece HOMEOFFICE Pow...	
10047	Supplies	Martin Yale Chadless Opener Electric Letter Op...	
10066	Chairs	Hon GuestStacker Chair	
10121	Copiers	Hewlett Packard LaserJet 3310 Copier	

	Sales Quantity	Discount	Profit	Profit_Ratio
5	2573.820	9	0.0 746.4078	0.290000
199	1579.746	7	0.4 -447.5947	-0.283333
224	1856.190	7	0.0 334.1142	0.180000
226	22638.480	6	0.5 -1811.0784	-0.080000
245	3499.930	7	0.0 909.9818	0.260000
...	...	...	...	...
9921	1669.600	4	0.0 116.8720	0.070000
10027	1704.560	13	0.0 511.3680	0.300000
10047	1665.620	2	0.0 33.3124	0.020000
10066	1586.690	7	0.0 412.5394	0.260000
10121	2879.952	6	0.2 1007.9832	0.350000

[251 rows x 22 columns]

```
In [19]:

# 9. Write a function to create a pivot table with Region as index, Category as columns, and Sales
# as sum.

def sales_pivot(df):
    return df.pivot_table(index='Region', columns='Category', values='Sales', aggfunc='sum')

print(sales_pivot(df))
```

```
stdout:
Category    Furniture  Office Supplies  Technology
Region
Central    164537.6518      168216.709  170416.312
East       212231.6960      211658.401  267938.071
South      117298.6840      125651.313  148771.908
West       260679.7295      226366.891  252766.988
```

```
In [20]:

# 10. Write a function to filter Furniture category, group by Sub-Category, return average Profit, and
# sort descending.

def furniture_avg_profit(df):
    return df[df['Category'] == 'Furniture'].groupby('Sub-Category')['Profit'].mean().sort_values(ascending=False)

print(furniture_avg_profit(df))
```

```
stdout:
Sub-Category
Chairs      42.939325
Furnishings  13.767833
Bookcases   -15.655490
Tables      -54.457687
Name: Profit, dtype: float64
```